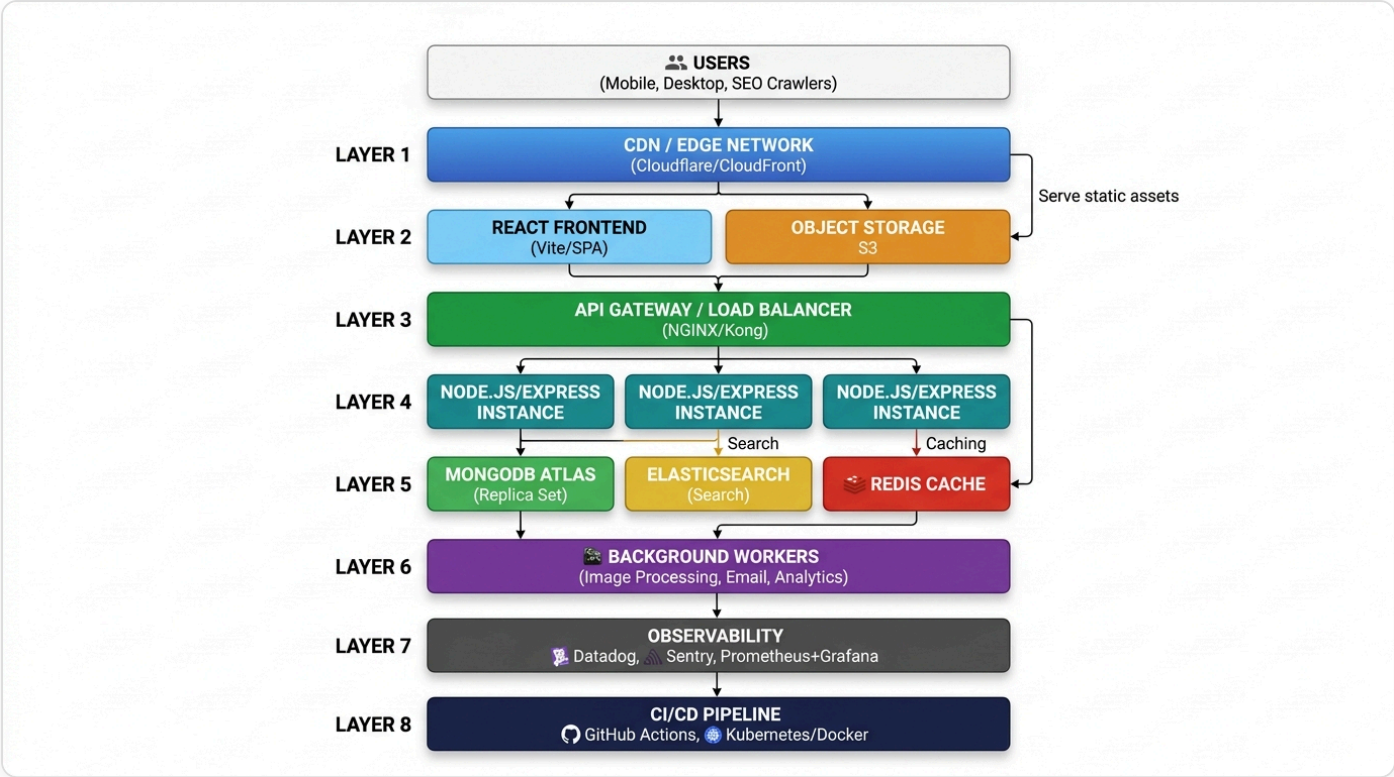


StayGallery — Production Architecture Specification

Scalable Cloud-Native Architecture for Vacation-Rental Marketplace

1. System Architecture Diagram



2. Architectural Layers & Technology Stack

2.1 Client Layer

- **React 18 SPA (Vite):** Fast client-side rendering with dynamic routing via React Router v6.
- **State Management:** React Context API (AuthContext, WishlistContext) providing reactive authentication and persistence.
- **Styling:** Scoped CSS Modules with custom design tokens for responsive, modular components.

2.2 CDN & Edge Tier

- **Cloudflare / AWS CloudFront:** Global Anycast Points of Presence delivering sub-30ms latency worldwide.
- **Asset Optimization:** Automatic on-the-fly WebP and AVIF image compression.
- **Security:** Distributed Denial-of-Service (DDoS) protection, Web Application Firewall (WAF), and SSL termination.

2.3 API Gateway & Load Balancing

- **AWS Application Load Balancer / NGINX Plus:** Intelligent path-based routing (/api/*), health check monitoring, and SSL offloading.
- **Rate Limiting:** Tiered rate limiting protecting authentication and write endpoints.

2.4 Application Microservice Cluster

- **Stateless Node.js + Express Services:** Horizontally auto-scaled containers on AWS ECS / Kubernetes.
- **Authentication:** Secure HTTP-only JWT cookies mitigating XSS vulnerabilities.
- **Asynchronous Workers:** BullMQ with Redis for background jobs, email confirmations, and media pipelines.

2.5 Persistence & In-Memory Data

- **MongoDB Atlas:** Multi-AZ replica set with primary/secondary automatic failover and ACID transactions.
- **Redis Cluster:** In-memory session store, query caching, and distributed locking (Redlock) preventing double bookings.
- **Elasticsearch:** High-performance geospatial and full-text search engine for instant listing filtering.
- **AWS S3 / Cloudflare R2:** Highly durable object storage for high-resolution property gallery media.